A Closed Pointed Cone Outside q$\mathcal{G}^*$ for Which the Bounded-Base Equivalence Holds

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid, for human review. It gives a full negative answer to Problem 1 of García-Castaño–Melguizo Padial, including under the natural restriction to closed pointed cones in Banach spaces.

1 Source problem

For a cone C in a normed space X, let

$$C^* = \{f \in X^* : f(c) \geq 0 \text{ for every } c \in C\}.$$

The family q$\mathcal{G}^*$ consists of the cones whose dual cone is quasi-generating, that is,

$$\overline{C^* - C^*}^{\|\cdot\|} = X^*.$$

García-Castaño and Melguizo Padial prove that if $C \in \mathrm{q}\mathcal{G}^*$, then

$$\left[0 \notin \overline{C \setminus \epsilon B_X}^{\text{weak}} \quad \text{for every } \epsilon \in (0,1)\right] \quad \Longleftrightarrow \quad [C \text{ has a bounded base}]. \tag{1}$$

Their Problem 1 asks whether q$\mathcal{G}^*$ is maximal among families of cones for which (1) holds [1]. Since their motivation explicitly notes that closed cones need not belong to q$\mathcal{G}^*$, it is useful to answer the question without exploiting a nonclosed or nonpointed cone.

The complete source page, including the definition of a base, Corollary 1, and Problem 1, follows.

refer the reader to [2]–. It is not difficult to prove that dentability of a cone at the origin implies normality of such a cone. On the other hand, by [1, Theorem 4.4] any normal cone belongs to q$\mathcal{G}^*$ and by [2, Lemma 2.39] any pointed cone in a normed vector lattice is normal.

Remark 2 The family q$\mathcal{G}^*$ contains the family of normal cones and so any arbitrary pointed cone in a normed vector lattice.

It is worth pointing out that vector lattices provide a useful unifying framework for many problems on convex programming (see [5]).

Now, we will pay attention to Gong's question in [12, Conclusions]. Gong asked if for any normed space X and any closed cone C the condition $0_X \notin \overline{C \setminus \epsilon B_X}^{\text{weak}}$, $\forall \epsilon \in (0,1)$ is really weaker than the cone C has a bounded base. Let us recall that a *base* for a cone C is a nonempty convex subset $B \subset C$ such that $0_X \notin \overline{B}$ and each $c \in C \setminus \{0_X\}$ has a unique representation of the form $c = \lambda b$ for some $\lambda > 0$ and $b \in B$. A base is called a *bounded base* if it is a bounded subset of X. Bounded bases have been widely studied and provide a useful tool in many topics such as in the theory of Pareto efficient points in [6], in reflexivity of Banach spaces in [7], and in density theorems in [12]. It is known that a cone C has a bounded base if and only if the origin is a denting point of C. In [11, Example 1.5] we provided a non closed cone for which the answer to Gong's question is positive. In this paper, we provide a wide family of cones for which the answer to Gong's question is negative. Applying (iii)⇒(i) of Theorem 2, we can state.

Corollary 1 *Let X be a normed space and $C \subset X$ a cone which belongs to* q$\mathcal{G}^*$*. Then, the condition $0_X \notin \overline{C \setminus \epsilon B_X}^{weak}$, $\forall \epsilon \in (0,1)$, is equivalent to the condition that the cone C has a bounded base.*

On the other hand, [1, Example 4.6] shows that the closedness of a cone C does not imply $C \in$ q$\mathcal{G}^*$ –even when X is a Banach space–. We state the following problem for a future research.

Problem 1 Is q$\mathcal{G}^*$ maximal among those families of cones for which Corollary 1 holds true?

Gong's question is directly related to the following two results on density of Arrow, Barankin and Blackwell's type: [20, Corollary 4.2] –due to Petschke– and [12, Theorem 3.2 (a)] –due to Gong–. Both results concern the approximation of the Pareto efficient points of compact convex subsets by points that are maximizers of some strictly positive functional on this set. Given a subset $A \subset X$, the set of *maximal (or efficient) points of* A (with respect to the cone C) is defined as $Max(A, C) := \{x \in A \colon \{x\} = A \cap (x + C)\}$. On the other hand, given a cone C the *set of all strictly positive functionals* is defined by $C^\# := \{f \in X^* \colon f(c) > 0, \forall c \in C, c \neq 0_X\}$, and the set of *positive (or proper efficient) points* of A as $Pos(A, C) := \{x \in A \colon \exists f \in C^\#, f(x) = \sup_A f\}$. It is straightforward that $Pos(A, C) \subset Max(A, C)$, however this inclusion is strict. Next, we state the density results of Petschke and Gong.

Theorem 5 *(Petschke) Let A be a weak compact convex subset of the normed space X and assume that C is a closed cone with a bounded base. Then $Max(A, C) \subset \overline{Pos(A, C)}^{\|\cdot\|}$.*

2 Proof intuition

An epigraph cone converts the geometry of an auxiliary norm into order geometry. We choose a continuous but much weaker norm p on c_0 with two special sequences. First, the usual unit vectors have ambient norm one but p-norm tending to zero; their lifts to the boundary of the epigraph are therefore weakly null points of norm one. This simultaneously destroys the weak separation condition and every bounded base. Second, the initial-block vectors $e_1 + \cdots + e_N$ have p-norm tending to zero. Positivity of a dual functional on both signs of these boundary vectors then forces the sum of its ℓ^1 coordinates to vanish. Thus the entire dual cone spans only a proper hyperplane. The epigraph remains closed and pointed because p is a genuine continuous norm.

3 Construction and proof

Work over the real field. Put

$$E = c_0(\mathbb{N}), \qquad X = E \oplus_\infty \mathbb{R}, \qquad \|(x,t)\| = \max\{\|x\|_\infty, |t|\}.$$

Define, for $x = (x_k)_{k\geq 1} \in c_0$,

$$p(x) := \sup_{k\geq 1} 2^{-k}|x_k - x_{k+1}|. \tag{2}$$

The function p is a norm on c_0: if all successive differences vanish, then x is constant, hence $x = 0$ because $x \in c_0$. Moreover $p(x) \leq \|x\|_\infty$, so p is continuous for the usual norm.

Consider its epigraph cone

$$C := \{(x,t) \in X : t \geq p(x)\}. \tag{3}$$

Theorem 1. *The cone C in* (3) *is closed and pointed, but $C \notin \mathrm{q}\mathcal{G}^*$. Nevertheless, the two assertions in* (1) *are both false for C. Consequently $\mathrm{q}\mathcal{G}^*$ is not maximal among families of cones for which* (1) *holds. This remains true if the ambient class is restricted to closed pointed cones in Banach spaces.*

Proof. Since p is continuous and convex, its epigraph C is a closed convex cone. If both (x,t) and $(-x,-t)$ belong to C, then

$$t \geq p(x) \quad \text{and} \quad -t \geq p(-x) = p(x).$$

Thus $p(x) = 0$ and $t = 0$, whence $x = 0$. Therefore C is pointed.

We next show that its dual cone is not quasi-generating. Identify $X^* = \ell^1 \oplus_1 \mathbb{R}$ and write its elements as (f,a), acting by

$$\langle (f,a),(x,t)\rangle = \sum_{k\geq 1} f_k x_k + at.$$

Positivity on $(0,t)$ for all $t \geq 0$ forces $a \geq 0$. Positivity on the boundary points $(x,p(x))$ and $(-x,p(x))$ then gives

$$\left|\sum_{k\geq 1} f_k x_k\right| \leq ap(x) \qquad (x \in c_0). \tag{4}$$

Conversely, $a \geq 0$ and (4) plainly suffice for $(f,a) \in C^*$.

Let $s_N = e_1 + \cdots + e_N$. Then $p(s_N) = 2^{-N}$, so (4) yields

$$\left|\sum_{k=1}^{N} f_k\right| \leq a2^{-N}.$$

Because $f \in \ell^1$, passage to the limit gives

$$\sum_{k \geq 1} f_k = 0. \tag{5}$$

Hence

$$C^* - C^* \subset \left\{ (f, a) \in \ell^1 \oplus_1 \mathbb{R} : \sum_{k \geq 1} f_k = 0 \right\}. \tag{6}$$

The right-hand side is a proper norm-closed hyperplane of X^*. Thus $C^* - C^*$ is not norm dense and $C \notin \mathrm{q}\mathcal{G}^*$.

It remains to test (1). For $n \geq 2$ let

$$u_n = (e_n, p(e_n)) \in C.$$

Here $p(e_n) = 2^{-(n-1)}$, so $p(e_n) \to 0$. The canonical unit vectors are weakly null in c_0, and therefore

$$u_n \rightharpoonup 0 \quad \text{in } X, \qquad \|u_n\| = 1. \tag{7}$$

For every $\epsilon \in (0, 1)$, the entire sequence lies in $C \setminus \epsilon B_X$. Equation (7) proves

$$0 \in \overline{C \setminus \epsilon B_X}^{\text{weak}} \qquad \text{for every } \epsilon \in (0, 1), \tag{8}$$

so the left assertion in (1) is false.

For completeness, (7) also directly rules out a bounded base. Indeed, suppose B were a bounded base for C. By the definition used in the source, B is convex and $0 \notin \overline{B}$. Strong separation gives $\Phi \in X^*$ and $\alpha > 0$ such that $\Phi(b) \geq \alpha$ for every $b \in B$. Put $M = \sup_{b \in B} \|b\| < \infty$. Every $c \in C \setminus \{0\}$ has the form $c = \lambda b$, and if $\|c\| \geq \epsilon$ then $\lambda \geq \epsilon / M$. Consequently

$$\Phi(c) \geq \frac{\alpha \epsilon}{M} \qquad (c \in C \setminus \epsilon B_X). \tag{9}$$

The weak neighborhood $\{x \in X : |\Phi(x)| < \alpha\epsilon/(2M)\}$ of zero is disjoint from the truncated cone, contradicting (8). Thus C has no bounded base, so the right assertion in (1) is false as well.

Both sides of (1) are false, hence their equivalence holds for C. Adjoining this single cone to $\mathrm{q}\mathcal{G}^*$ produces a strictly larger family on which (1) holds, which proves the maximality claim false. $\square$

4 Scope

The example does not merely exploit the fact that the source definition of a cone allows nonpointed or nonclosed sets. It is a closed pointed cone in the Banach space $c_0 \oplus_\infty \mathbb{R}$, precisely the setting suggested by the sentence immediately preceding Problem 1. The conclusion is the literal set-theoretic nonmaximality asked there. It does not propose a maximal *natural* or structurally defined replacement for $\mathrm{q}\mathcal{G}^*$.

5 Verification and novelty scope

Every substantive step is contained in (2)–(9). In particular, the dual obstruction is witnessed by the explicit continuous functional $(f, a) \mapsto \sum_k f_k$ on X^*, and the bounded-base obstruction is witnessed by an explicit weakly null sequence of norm one.

A bounded literature audit was performed on 9 August 2026 after the direct construction. The run indexes, exact-phrase and core-keyword web searches, the current arXiv record, the 2019 journal version, and later papers citing the source were checked. No later resolution of Problem 1 or occurrence of this epigraph-cone example was found. This supports plausible novelty only; it does not certify priority.

References

[1] F. García-Castaño and M. A. Melguizo Padial, *On dentability and cones with a large dual*, Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. RACSAM **113** (2019), 2679–2690; doi:10.1007/s13398-019-00650-3, also arXiv:2401.10102.